

Host-Based Intrusion Detection System (HIDS)

Rule Documentation & Detection Guide

ProStackHub HIDS + AI Log Chatbot

Purpose

This document describes the rule-based detection logic used by the HIDS project to identify suspicious events in monitored security logs. The rules are designed for defensive monitoring and lab validation.

Detection pipeline

Log file -> Watchdog real-time monitoring -> Regex/custom rules -> Alert classification -> SQLite storage -> Flask dashboard -> Groq AI analysis

Project status

Capability	Status
Log monitoring	Implemented
Regex detection rules	Implemented
SQLite alert storage	Implemented
Flask dashboard	Implemented
Groq AI analysis	Implemented / tested directly
Nmap validation	Implemented
Hydra validation	Pending / lab validation

Detection Rules

R01 - Failed Login

Purpose: Detect repeated or explicit failed authentication messages.

Severity: MEDIUM

Patterns: failed login; authentication failure; login failed; invalid password

Recommended response: Investigate account activity, verify the source IP, and review authentication logs.

R02 - Brute Force Attack

Purpose: Detect a burst of failed logins from the same source within the configured time window.

Severity: HIGH

Patterns: multiple failed login; brute force; too many authentication failures

Recommended response: Block or rate-limit the source when appropriate, reset affected credentials, and review successful logins after the attempts.

R03 - Privilege Escalation

Purpose: Detect suspicious privilege-related activity.

Severity: HIGH

Patterns: privilege escalation; root access; administrator privilege; sudo

Recommended response: Verify whether the elevation was authorized, review the account and process involved, and remove unauthorized privileges.

R04 - Port Scanning

Purpose: Detect evidence of port scanning or multiple-port probing.

Severity: MEDIUM

Patterns: port scan; nmap scan; scanning ports; multiple ports

Recommended response: Identify the scanner source, confirm whether the activity is authorized, and review exposed services.

Brute-Force Detection Logic

The detector tracks failed-login timestamps by source IP. When the number of failed attempts reaches the configured threshold within the configured time window, the system creates a HIGH-severity Brute Force Attack alert.

Current configuration

Failed-login threshold: 5 attempts

Observation window: 60 seconds

Event severity: HIGH

Example test log

```
2026-08-22 19:00:01 FAILED LOGIN user=admin ip=127.0.0.2
2026-08-22 19:00:02 FAILED LOGIN user=admin ip=127.0.0.2
2026-08-22 19:00:03 FAILED LOGIN user=admin ip=127.0.0.2
2026-08-22 19:00:04 FAILED LOGIN user=admin ip=127.0.0.2
2026-08-22 19:00:05 FAILED LOGIN user=admin ip=127.0.0.2
```

Expected result

The HIDS records multiple Failed Login events and then raises a Brute Force Attack alert for the same source IP. The alert is stored in SQLite and displayed on the dashboard.

Port-Scan Validation

The project includes a controlled Nmap validation script that scans the local host (127.0.0.1) and writes a corresponding port-scan event into the monitored log for HIDS validation.

Expected detection

Event: Port Scanning

Severity: MEDIUM

Source: 127.0.0.1 (local validation)

Alert Storage & Dashboard

Detected events are stored in SQLite with timestamp, event type, severity, source IP, and original log message. The Flask dashboard reads these records and presents summary counters and a detailed event table.

Field	Description
timestamp	Time the alert was stored
event_type	Detection rule that matched
severity	LOW / MEDIUM / HIGH / CRITICAL
source_ip	Source address extracted from the log when available
log_message	Original log line used for the detection

AI Analysis Layer

The Groq-powered chatbot reads recent HIDS alerts and provides defensive analysis, including what happened, the relevant event, severity/risk, possible attack technique, and remediation guidance. The chatbot is instructed not to invent events that are not present in the supplied alerts.

Security note: the Groq API key must remain in the local .env file and must never be committed to GitHub.

Testing & Validation Checklist

Test	Expected outcome
Failed login	MEDIUM Failed Login alert appears
Five failed logins from one IP within 60 seconds	HIGH Brute Force Attack alert appears
Privilege escalation log	HIGH Privilege Escalation alert appears
Nmap local validation	MEDIUM Port Scanning alert appears
Dashboard refresh	Alerts and severity counters update
Groq direct test	AI returns a response
Hydra validation	Pending; use only against an authorized local lab target

Operational Safety

Use the HIDS only on systems and logs you own or are explicitly authorized to monitor. For Nmap/Hydra validation, use localhost or a controlled lab environment. Keep API keys and other secrets out of source control.

End of document